INCH-POUND
MIL-C-83500(USAF)
AMENDMENT 3
23 February 1994
SUPERSEDING
AMENDMENT 2
16 June 1980

MILITARY SPECIFICATION

CAPACITORS, FIXED, ELECTROLYTIC, (NONSOLID ELECTROLYTE),
TANTALUM ANODE AND CATHODE,
GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-C-83500(USAF), dated 19 June 1975, and is approved for use the Department of the of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title, delete and substitute as printed above.

1.1, line 1, following "Tantalum" add "anode and".

* 2.1, SPECIFICATIONS, MILITARY, delete "MIL-C-45662" and its title.

* 2.1, STANDARDS, MILITARY, delete: "MIL-STD-105" and its title.

* 2.1, STANDARDS, MILITARY, add: "MIL-STD-45662 - Calibration Systems Requirements."

* Bottom of page, add the following:

"

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Air Force Material Command, 645 LOG/LGME, Gentile Station, 1060 Hamilton Street, Dayton, OH 45444-5400 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

"

PAGE 11

* TABLE II, delete and substitute the following:

" TABLE II. Group A inspection.

Examination or test	Requirement paragraph	Test paragraph	Sampling procedure
Burn-in	3.5	4.6.2	100% inspection
Visual and mechanical examination (external)			
Materials	3.1 and 3.3	4.6.1	13 samples 0 failures
Physical dimensions	3.1	4.6.1	
Marking 1/	3.29	4.6.1	
Workmanship	3.30	4.6.1	
DC leakage	3.6	4.6.3	100% inspection
Capacitance	3.7	4.6.4	
Dissipation factor	3.8	4.6.5	

1/ Marking defects shall be based on visual examination only. Any subsequent electrical defects shall not be used as a basis for determining marking defects."

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:18Z
Check the source to verify that this is the current version before use.

* 4.5.1.2.1, delete in its entirety.

TABLE III, Subgroup 2, before "Life at 85°C (240-hour)", add:

Test	Requirement paragraph	Test paragraph
Thermal shock 1/	3.11	4.6.8

1/ Twenty five cycles."

PAGE 12

* 4.5.1.3.1, delete and substitute the following:

"4.5.1.3.1 Sampling plan. For subgroup 1, 13 samples shall be selected from units which have passed group A inspection; no failures permitted. For subgroup 2, 10 samples shall be selected from units which have passed group A inspection; one failure is permitted."

PAGE 13

TABLE IV, Subgroup 5, delete "every 4 months" and substitute "every 12 months".

TABLE IV, Subgroup 6, delete and substitute the following:

Test	Requirement paragraph	Test paragraph
Barometric pressure (reduced) Life at 85°C	3.20 3.22.3	4.6.17 4.6.19.3

PAGE 14

4.6.7(a), delete and substitute:

"(a) Mounting of specimens - External mounting means."

PAGE 15

4.6.8(c), delete and substitute:

"(c) Number of cycles - 300 (except group B inspection - 25 cycles)."

PAGE 17

TABLE V, title, delete and substitute: "DC surge voltages, derated surge voltages, and rated voltages."

TABLE V, Column 2: Delete "Derated voltage at 125°C" and substitute "Derated surge voltage at 125°C."

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:18Z
Check the source to verify that this is the current version before use.

PAGE 18

4.6.15, delete and substitute the following:

"4.6.15 Seal (see 3.18). Capacitors shall be tested in accordance with method 112 of MIL-STD-202. The following details and exceptions shall apply:

(a) Test condition - C.

(b) Degree of leakage rate sensitivity: 10^{-8} atm cc/s.

(c) Procedure: The mass-spectrometer-type leak detector shall be coupled to the unsealed evacuation port of the specimen, and a vacuum created within the specimen. It is extremely important that the coupling connections between the specimen and the leak detector be perfectly sealed. The specimen shall then be subjected to a gas or by spraying the specimen thoroughly with a jet of the gas. If a defect exists in the specimen, an amount of gas that depends upon the size of the defect will be drawn through it and passed into the leak detector, which will read the leakage rate.

(d) Measurements after test: Not applicable.

Following this test, capacitors shall be visually examined for evidence of leakage."

PAGE 20

4.6.20(b), delete and substitute the following:

"(b) Operating conditions-derated voltage (see figure 2) shall be applied."

PAGE 21

4.6.22, delete and substitute the following:

"4.6.22 AC ripple life at 125°C (see 3.25). Capacitors shall be tested as specified in 4.6.19.1, with the following exceptions:

(a) Test temperature and tolerance: 125°C +4°C, -0°C.

(b) Operating conditions: Capacitors shall be tested at 40 ±2 kHz sinusoidal and 750 mA rms (±10 percent). The peak ac voltage plus dc shall not exceed the rated voltage of the capacitor."

PAGE 22

* 6.3, delete and substitute the following:

"6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List (QPL) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Defense Electronics Supply Center (DESC-EQP), 1507 Wilmington Pike, Dayton, Ohio 45444-5287. agent for information pertaining to qualification of products. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3.1)."

6.3.1 Copies of SD-6. Copies of SD-6, "Provisions Governing Qualification" may be obtained upon application to Defense Printing Service Detachment Office, Bldg. 40 (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094."

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:18Z
Check the source to verify that this is the current version before use.

Add the following new paragraph:

"6.5 Mounting of units. Mounting of these units will be left to the discretion of the user. Thixotropic compound or other mechanical clamping may be used."

PAGE 23

10.1, delete and substitute the following:

"10.1 Scope. This appendix details the qualification submittal plans for capacitors to be subjected to the qualification inspection specified in this specification. These plans specify the sample size, sampling criteria, and limits of coverage for the type of qualification sought."

10.2 Qualification categories. Delete in its entirety. (This paragraph was added by Amendment 1.)

PAGE 24

TABLE VIII, delete and substitute the following:

" TABLE VIII. Combined-style submittal data.

Style	Part number	Sample size	Rated voltage	Voltage group
CRL01	M83500/01-0001	9	6	I
	M83500/01-1001	9	6	
	M83500/01-0015	9	30	
	M83500/01-1015	9	30	
CRL02	M83500/01-0021	9	50	II
	M83500/01-1021	9	50	
	M83500/01-0033	9	125	
	M83500/01-1033	9	125	

"

" End of document, DD Form 1426 address, delete and substitute the following:

"645 LOG/LGME
Gentile Station
1060 Hamilton Street
Dayton, OH 45444-5400".

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodian:
Air Force - 85

Review activity:
Air Force - 99

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5910-F475)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:18Z
Check the source to verify that this is the current version before use.